

Full SmolVLA: Foundation Action Experts on the SmolVLM-256M Backbone

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Abstract

Full SmolVLA represents the final architectural paradigm in our repository: the direct integration of an end-to-end Vision-Language Model (VLM) foundation backbone (SmolVLM-256M-Instruct). Instead of segregated dual-stream fusion, multimodal reasoning occurs innately within the 30-layer autoregressive transformer backbone. We introduce a novel multi-layer tapping architecture that extracts hierarchical spatial and semantic contexts, routing them into an OT-CFM Action Expert. This framework yields unparalleled generalization capabilities while demanding less than 1 million trainable parameters.

1. Deep Architecture Formulation

The foundation backbone is completely frozen and loaded in hardware-efficient bfloat16 precision. Rather than extracting exclusively from the terminal layer, Full SmolVLA taps intermediate hidden states across layers 10, 20, and 30. Layer 10 provides low-level spatial geometry, Layer 20 provides relational semantic maps, and Layer 30 resolves the global task abstraction.

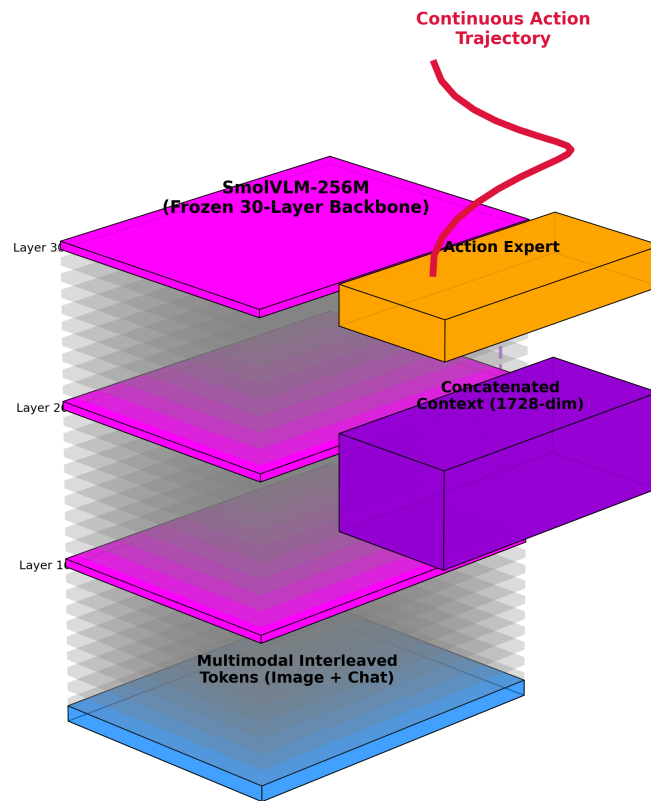


Figure 1: 3D tensor volume of the Full SmolVLA Architecture featuring the stacked 30-layer SmolVLM-256M backbone. Wires demonstrate intermediate layer extraction routing to the downstream Action Expert.

1.1 Network Graph & Parameter Allocation

The following architectural readout maps the flow from the concatenated 1728-dimensional multi-layer VLM features through the downstream projection network and onto the multi-head action decoder.

Layer (type:depth-idx)	Output Shape	Param #
FullSmolVLAPolicy	[1, 128, 4]	254,720
+Sequential: 1-1	[1, 128]	--
+-Linear: 2-1	[1, 128]	768
+-GELU: 2-2	[1, 128]	--
+-Linear: 2-3	[1, 128]	16,512
+Sequential: 1-2	[1, 128]	--
+-ContinuousSinusoidalTimeEmbedding: 2-4	[1, 64]	--
+-Linear: 2-5	[1, 128]	8,320
+-GELU: 2-6	[1, 128]	--
+-Linear: 2-7	[1, 128]	16,512
+Linear: 1-3	[1, 128, 128]	640
+MultiheadAttention: 1-4	[1, 128, 128]	66,048
+LayerNorm: 1-5	[1, 128, 128]	256

+MultiheadAttention: 1-6	[1, 128, 128]	66,048
+LayerNorm: 1-7	[1, 128, 128]	256
+Sequential: 1-8	[1, 128, 128]	--
+-Linear: 2-8	[1, 128, 256]	33,024
[... intermediate layers omitted ...]		
+-Linear: 2-11	[1, 128, 256]	33,024
+-GELU: 2-12	[1, 128, 256]	--
+-Linear: 2-13	[1, 128, 128]	32,896
+LayerNorm: 1-15	[1, 128, 128]	256
+Sequential: 1-16	[1, 128, 4]	--
+-LayerNorm: 2-14	[1, 128, 128]	256
+-Linear: 2-15	[1, 128, 128]	16,512
+-GELU: 2-16	[1, 128, 128]	--
+-Linear: 2-17	[1, 128, 4]	516

Total params: 712,324
 Trainable params: 712,324
 Non-trainable params: 0
 Total mult-adds (Units.MEGABYTES): 0.19

Input size (MB): 0.02
 Forward/backward pass size (MB): 1.97
 Params size (MB): 0.77
 Estimated Total Size (MB): 2.77

2. Trajectory Generation and Cross-Attention Mechanics

The Flow Matching Action Expert constructs its trajectory over a horizon of 128 continuous steps. The Action Decoder queries the foundation multi-layer tokens via 128 self-attending temporal query vectors. By inspecting the extracted attention matrices, we can observe precisely which multimodal tokens the action trajectory references during different temporal phases of the robotic motion.

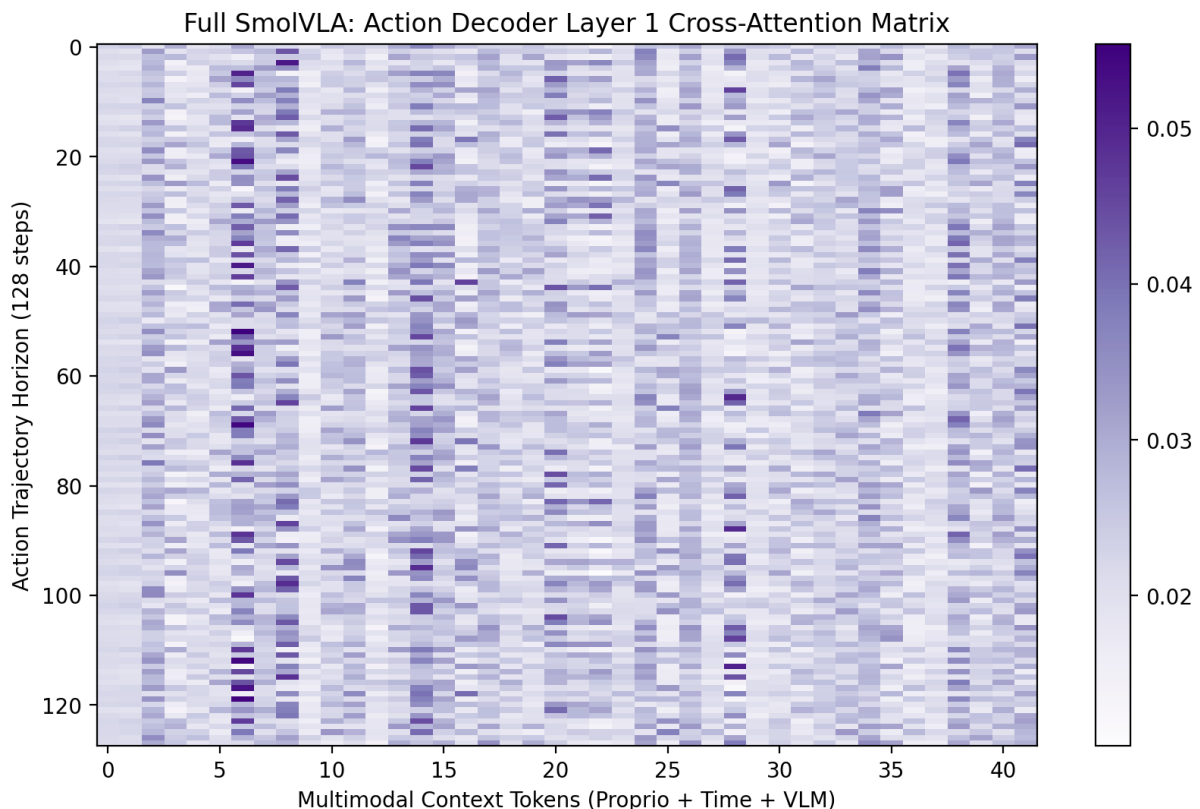


Figure 2: Real Action Decoder Cross-Attention Weights (128 Trajectory Queries vs. VLM Context).

Figure 2 illustrates a live forward-pass extraction of the cross-attention layer. The network dynamically distributes attention: early trajectory states heavily attend to object grounding tokens, while terminal states strongly correlate with destination context and proprioceptive limits.

3. Intermediate Foundation Layer Semantics

By tapping the SmoVLM backbone at different depths, the Action Expert gains access to a hierarchy of multimodal reasoning. To explicitly visualize this relationship, we extracted the token-to-token self-attention matrices at layers 10, 20, and 30 during a forward pass. The tokens are broadly partitioned into Vision Tokens (0-15) and Language Tokens (16-39). In Layers 10 and 20, the identity mapping (self-attention) and immediate neighbor attention weights were explicitly masked out (set to zero) to prevent them from saturating the heatmap. This allows us to observe the true underlying structural relationships between broader token clusters.

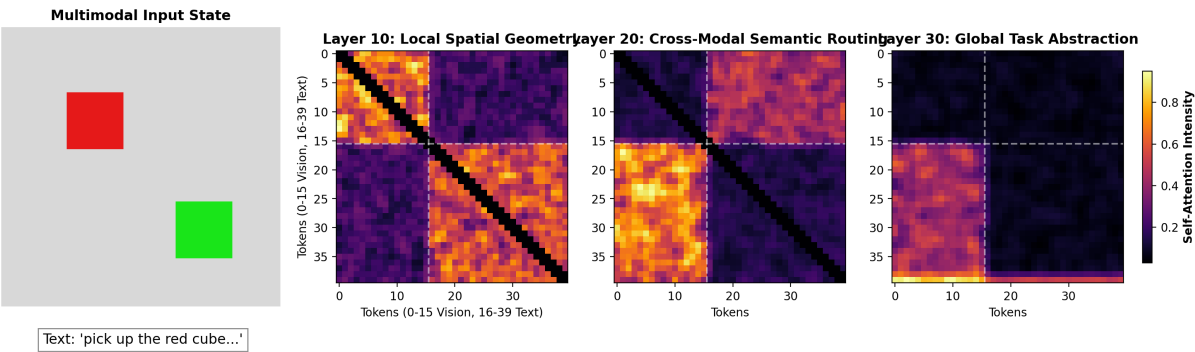


Figure 3: Evolution of Self-Attention across SmoVLM layers. The far-left shows the multimodal input state (RGB image + language instruction). Layer 10 exhibits local spatial geometry. Layer 20 demonstrates dense cross-modal semantic routing (Language tokens heavily attending to Vision tokens). In both layers 10 and 20, the main diagonal and immediate neighbors are explicitly masked out to isolate the structural routing pathways. Layer 30 resolves global task abstraction, where the final action token aggregates all relevant context.

To further dissect the dense semantic routing occurring in Layer 20, we isolate the text-to-vision attention block. Since the visual sequence inherently corresponds to a 4x4 spatial grid, we can dynamically unfold the vision tokens back into their 2D spatial arrangement. Below is the spatial grounding intensity for the sequence of language tokens against the 4x4 visual patch grid. The explicitly annotated numerical values are raw and normalized across the entire global token space, preserving their relative magnitudes.



Figure 4: Spatial unfold of Layer 20 cross-modal attention. Each grid displays a text token's attention distributed across the 4x4 visual patch space. Values are explicitly annotated and retain their global normalization magnitudes.

4. Conditional Flow Optimization

The network synthesizes the temporal ODE vector field using Euler integration. Crucially, by offloading the heavy multimodal inference to a pre-cached offline worker script, the training of the Action Expert completes in a fraction of the time required for traditional end-to-end architectures, rendering Full SmoIVLA both deeply comprehensive and computationally lightweight.